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Mapping the Semantic Frontier of Bangla (1950–2025): A Hybrid N-gram and Procrustes-Aligned Embedding Analysis of Diachronic Lexical Change

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Languages are not static systems: the meanings of words shift over time in response to technological, political, and cultural change. Detecting and quantifying this Lexical Semantic Change (LSC) is therefore crucial for building temporally-aware Natural Language Processing (NLP) systems and for advancing our understanding of how societies evolve through language. Despite Bangla being the seventh most spoken language in the world, it remains entirely unstudied from a computational diachronic perspective—no prior work has applied modern LSC detection methods to the Bangla lexicon. This gap is primarily attributable to the absence of a structured, time-stratified corpus and the severe noise introduced by Optical Character Recognition (OCR) of degraded historical print archives.

This paper addresses both obstacles by presenting the first large-scale, data-driven analysis of lexical semantic change in Bangla across a pivotal 75-year window spanning 1950 to 2025. We construct a four-era diachronic corpus from newspapers and literary works and introduce an error-aware preprocessing pipeline specifically designed to mitigate the approximately 20% character error rate observed in our OCR output from mid-20th-century archives. To detect and quantify semantic change, we apply a hybrid analytical pipeline that combines two complementary signals: (i) an *N-gram Module* using smoothed frequency trajectories and Positive Pointwise Mutual Information (PPMI) collocate analysis to capture transparent, interpretable surface-level changes; and (ii) an *Embedding Module* employing per-era Word2Vec (Skip-gram) models aligned via Orthogonal Procrustes transformation to quantify latent semantic drift. A unified *ChangeScore* synthesises both signals into a single ranked list of the most dynamic words in the Bangla lexicon.

Qualitative case studies on three representative words— (*digital*), (*kul/cool*), and (*independence*)—illustrate three distinct patterns of change: lexical emergence tied to technological adoption, semantic replacement driven by cultural globalisation, and meaning anchoring shaped by national history. Quantitatively, our analysis confirms that the Law of Frequency—the cross-linguistic principle that more frequent words change more slowly—holds for Bangla with a statistically significant negative Spearman correlation ($\rho = -0.083$, $p < 0.001$). We do not find evidence for the Law of Polysemy under our corpus conditions, a null result we attribute to data sparsity in the early eras rather than to a genuine linguistic exception.

To the best of our knowledge, this is the first systematic computational study of lexical semantic change in Bangla, and the first application of Orthogonal Procrustes alignment and PPMI collocate analysis to construct a longitudinal map of this language’s evolution. The diachronic corpus and the analysis pipeline are released to support future work on Bangla and other low-resource Indo-Aryan languages.

CCS Concepts: • **Applied computing** → **Linguistics**; • **Computing methodologies** → **Natural language processing**; • **Information systems** → *Data mining*.

Additional Key Words and Phrases: Lexical semantic change, diachronic NLP, Bangla NLP, word embeddings, Orthogonal Procrustes, PPMI, low-resource languages, computational linguistics, language evolution

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1 Introduction

Language is not a fixed code but a living system whose meanings are continuously renegotiated by the communities that use it. Over decades, existing words acquire new senses, lose old ones, or shift their connotation entirely in response to technological innovation, political upheaval, and cultural exchange [29]. This process—termed *Lexical Semantic Change* (LSC)—poses a fundamental challenge to Natural Language Processing (NLP): a system trained on contemporary text will systematically misinterpret historical documents if it treats word meanings as static [10]. Addressing this challenge requires both a computational framework capable of detecting and quantifying meaning shift and a time-stratified corpus that faithfully reflects the language across the period of interest.

Bangla (Bengali) presents a compelling yet under-explored case for LSC research. As the seventh most spoken language in the world with approximately 250 million speakers [27], Bangla has undergone profound semantic disruption over the past 75 years. The transition from colonial rule, the liberation of Bangladesh in 1971, rapid urbanisation, and the global digital revolution have all deposited their traces in the lexicon—shifting the meaning of political terms, naturalising foreign loanwords, and generating an entirely new vocabulary for technology. Despite this rich potential, Bangla remains a *computational diachronic blank*: to the best of our knowledge, no prior work has applied modern LSC detection methods to Bangla text spanning multiple historical periods. This gap stands in sharp contrast to the depth of diachronic research available for high-resource languages such as English, German, and Swedish, for which shared evaluation benchmarks, annotated datasets, and multiple competing systems already exist [24].

The absence of Bangla diachronic NLP is explained by two compounding obstacles. First, there is no publicly available, time-stratified Bangla corpus that provides sufficient coverage of the mid-20th century. Large-scale synchronic resources such as the KUMono 350-million word corpus [1] and the VACASPATI literary corpus [4] have greatly expanded the data landscape, but they lack the decade-by-decade temporal labelling required for diachronic study. Second, constructing such a corpus from historical archives requires Optical Character Recognition (OCR) of degraded physical documents—a process that introduces severe noise through character misrecognition, particularly problematic for Bangla’s complex ligature system and connected diacritical marks. Standard LSC pipelines assume clean, digital-first text and are not designed to be robust to character-level error rates in the range of 20%, which is what our own OCR experiments on mid-20th-century Bangla newsprint yield. This “noise-blindness” gap in existing methodology is a primary roadblock to achieving statistically reliable diachronic analysis for Bangla and other languages whose historical record lives in degraded print archives.

A further methodological limitation in the literature is its fragmentation between two distinct paradigms that rarely communicate with one another. Studies that use frequency-based n-gram signals to detect lexical emergence (neologisms and loanwords) operate independently of studies that use distributional vector drift to detect meaning shift in existing words [14, 18]. This separation produces an incomplete picture of linguistic dynamics, because the birth of new concepts and the re-imagination of old ones are complementary facets of the same evolutionary process. A unified framework that interleaves both signals is needed to produce a comprehensive “Dynamics Rank” of the Bangla lexicon.

This paper addresses all three gaps—data, noise robustness, and methodological integration—through the following contributions:

- (1) **A four-era diachronic Bangla corpus (1950–2025).** We compile the first time-stratified Bangla corpus spanning roughly eight decades, assembled from newspapers and literary texts and partitioned into four eras: 1950–1970, 1970–1990, 1990–2010, and 2010–2025. For older text (1950–2010) we apply an enhanced OCR

pipeline with multi-stage image preprocessing and post-correction, achieving sufficient quality for statistical and embedding analysis despite approximately 20% raw character error rate (CER).

- (2) **An error-aware hybrid analysis pipeline.** We introduce a two-module pipeline—an *N-gram Module* for transparent surface-level signals and an *Embedding Module* for latent semantic drift—whose individual outputs are fused into a single unified ChangeScore. The pipeline is explicitly designed to be robust to OCR noise through heuristic token filtering, frequency thresholding, and the use of subword-aware static embeddings.
- (3) **The first data-driven linguistic analysis of Bangla semantic evolution.** We present quantitative rankings of Bangla’s most dynamically changing words, three detailed qualitative case studies, and the first empirical test of the universal laws of semantic change [10] on a Bangla corpus.

The remainder of the paper is organised as follows. Section 2 reviews the evolution of LSC methodology and maps the specific gap in Bangla diachronic research. Section 3 describes the corpus construction strategy and the OCR pipeline. Section 4 formalises the hybrid analysis pipeline. Section 5 presents quantitative rankings, qualitative case studies, and the linguistic law tests. Section 6 interprets the findings, acknowledges limitations, and outlines future directions. Section 7 concludes.

2 Related Work

The computational detection of LSC has matured from manual philological observation to large-scale automated analysis over the past decade [30]. We organise the literature along three dimensions that directly motivate our methodological choices: (i) the evolution from count-based to embedding-based representations, (ii) the debate over static versus contextual embeddings, and (iii) the persistent low-resource gap that Bangla exemplifies.

2.1 From Count-Based Methods to Static Embeddings

Early computational work on semantic change relied on distributional co-occurrence counts, representing each word as a vector of association strengths with its context vocabulary. Gulordava and Baroni [9] demonstrated that comparing co-occurrence vectors trained on different time slices of the Google Books Ngram corpus could detect known shifts in English—for example, the drift of *gay* from meaning “cheerful” to referring to homosexuality—providing the first large-scale validation that distributional signals carry diachronic information. However, such count-based representations are inherently high-dimensional and sparse, and their comparison across time periods is methodologically straightforward only when the same context vocabulary defines the vector axes in both periods (as in Positive Pointwise Mutual Information (PPMI) matrices), because PPMI vectors share a coordinate system across corpora by construction.

The pivotal advance came from Hamilton et al. [10], who formalised a principled methodology for comparing *low-dimensional* distributional embeddings across time. The core problem they solved is this: models such as Word2Vec [17] and GloVe learn embedding spaces whose coordinate axes are arbitrary—two independently trained spaces will be mathematically incomparable even if the underlying language is identical. Hamilton et al. addressed this by applying *Orthogonal Procrustes* alignment [26]: an efficient singular-value-decomposition procedure that finds the rotation matrix $\mathbf{R}^*$ minimising the Frobenius norm between two aligned embedding matrices (Equation 4) while preserving all pairwise cosine similarities. After alignment, the cosine distance between a word’s vector in era t and era $t + \Delta$ serves as a direct, interpretable measure of how far its meaning has drifted. Using this methodology on six historical corpora spanning English, German, French, and Chinese, Hamilton et al. proposed two quantitative laws: the *Law of Conformity* (frequent words change more slowly) and the *Law of Innovation* (polysemous words change faster, independent

of frequency). These laws have since become the canonical benchmarks against which any new LSC study is evaluated [14, 29].

The SGNS+OP+CD pipeline—Skip-Gram with Negative Sampling (SGNS), Orthogonal Procrustes (OP) alignment, and Cosine Distance (CD) as the change score—was subsequently formalised by Schlechtweg et al. [23] as the dominant competitive baseline for LSC detection. Its robustness was further confirmed by Kaiser et al. [11], who won the Italian DIACR-Ita shared task with 94% accuracy using precisely this pipeline, outperforming BERT-based systems. This evidence motivates our choice of SGNS+OP as the backbone of our Embedding Module.

An important methodological critique was raised by Dubossarsky et al. [7], who showed that the frequency and polysemy correlations reported by Hamilton et al. are partially artifacts of the word representation models themselves rather than purely linguistic phenomena. This result does not invalidate the methodology but underscores the need for large, balanced corpora and careful controls—a challenge we address explicitly in our low-resource setting.

2.2 Contextual Embeddings and the Static/Contextual Debate

The introduction of contextual language models such as ELMo [19] and BERT [6] opened a new avenue for LSC detection. Unlike static embeddings, which assign a single vector to each word regardless of context, contextual models generate a distinct representation for each token occurrence. This distinction is crucial for polysemous words: when a word acquires a new sense while retaining its old one, a static embedding averages across both senses and may understate the shift, whereas contextual representations can distinguish the two sense distributions.

Giulianelli et al. [8] were the first to operationalise this advantage systematically. They extracted BERT representations for all instances of a target word from two time periods, clustered them into usage types, and measured semantic change via the Jensen-Shannon divergence between the two cluster distributions. Applied to English and German data, this approach detected subtle sense additions that static methods missed. Kutuzov and Giulianelli [13] extended the contextual approach to the SemEval-2020 Task 1 benchmark [24], achieving the best Subtask 2 results in post-evaluation using multilingual BERT. Martinc et al. [16] proposed fine-tuning ELMo separately on old and new corpora to obtain “historical” versus “modern” representations, finding this effective for detecting meaning differences even in morphologically rich languages.

Despite these promising results, contextual models have not consistently outperformed static methods. Laicher et al. [15] demonstrated that BERT does not outperform SGNS on the Italian DIACR-Ita shared task, and broader surveys confirm that the advantage of contextual models is data-dependent: they require sufficiently large per-era corpora to learn reliable sense distributions [18]. For the low-resource, OCR-noisy setting of our Bangla study, static Word2Vec embeddings are therefore the more appropriate and defensible choice. We explicitly discuss BanglaBERT [3] as a viable direction for future work once larger per-era Bangla corpora are available.

2.3 The SemEval-2020 Benchmark and Evaluation Standards

The community’s shared understanding of what constitutes rigorous LSC evaluation crystallised around SemEval-2020 Task 1 [24], which provided gold-standard datasets for English, German, Latin, and Swedish with human-annotated “changed” and “unchanged” word lists. Of 33 participating teams, the best system for graded change detection combined static and contextual embeddings [20], while the best binary classification system relied entirely on static Procrustes-aligned SGNS embeddings [22]. A parallel shared task in Spanish, LSCDiscovery [32], confirmed these

findings for a non-Germanic language. These evaluations are critical for contextualising our Bangla study: we follow the same general framework (per-era embedding training, alignment, cosine-distance scoring) but adapt it to the specific challenges of a South Asian, morphologically rich, low-resource language.

2.4 The Low-Resource Gap and the Case of Bangla

The entire body of work reviewed above was developed on, and validated for, a handful of high-resource, mostly Indo-European languages. A 2024 survey of LSC methods [18] explicitly notes that existing benchmarks cover only approximately 12 languages, all with substantial digital-first historical archives. The DHPLT resource paper [12] constructed diachronic corpora for 41 languages but identified Bangla as conspicuously absent. This is not merely a data gap but a theoretical gap: the Laws of Frequency and Polysemy have never been tested on an Indo-Aryan language characterised by complex inflectional morphology and extensive borrowing from Arabic, Persian, and English. It is therefore an open empirical question whether these laws are truly universal or whether they are artifacts of the Germanic and Romance language families in which they were discovered.

The Bangla NLP landscape has expanded substantially in recent years with synchronic resources and models. BanglaBERT [3] provides a state-of-the-art encoder pretrained on 27.5 GB of Bangla text; the KUMono 350-million word corpus [1] and the VACASPATI literary corpus [4] provide unprecedented scale for synchronic tasks. Reviews of Bangla NLP [2, 27] document active progress in sentiment analysis, named entity recognition, and machine translation. However, all these contributions are *synchronic*—they treat the language as a snapshot rather than as a trajectory. The only prior work adjacent to diachronic analysis is a study of Bengali neologisms during the COVID-19 pandemic [28], which used frequency-based methods to detect newly coined vocabulary in real time but did not address meaning shift over multi-decade spans. Our work is therefore the first to construct a multi-era diachronic corpus for Bangla and the first to apply aligned embedding methods to trace its semantic evolution.

3 Corpus Construction

To investigate semantic change over 75 years, we require text distributed across time in a way that reflects actual language use in each era. This section describes our corpus sources, our OCR pipeline for historical documents, and the preprocessing steps applied to all collected text.

3.1 Data Sources and Era Partitioning

We partition the diachronic corpus into four eras that correspond to major socio-political transitions in Bangla-speaking society:

- **Era 1 (1950–1970):** Post-Partition reconstruction and the language movement period.
- **Era 2 (1970–1990):** The Bangladesh Liberation War (1971) and the consolidation of an independent Bangla literary and journalistic tradition.
- **Era 3 (1990–2010):** Democratisation, economic liberalisation, and the early adoption of digital media.
- **Era 4 (2010–2025):** The smartphone era, social media, and the mainstreaming of English loanwords.

For Era 4 (2010–2025), we collected text from two sources: (i) a publicly available Bangla text dataset hosted on Kaggle containing web-crawled content, and (ii) large-scale web scraping of major online Bangladeshi news portals, including *Prothom Alo* and *BD-Pratidin*. We also incorporated digital copies of contemporary books available in PDF or ePub format. These sources provide clean, Unicode-encoded text and require no OCR processing.

For Eras 1–3 (1950–2010), we sourced text from scanned PDF archives of Bangla newspapers and printed books obtained from libraries and digital archives. Because no clean digital version of mid-20th-century Bangla periodicals exists, we processed these materials through an OCR pipeline described below. We ensured geographic balance by sourcing material from both Bangladesh (Dhaka-based publications) and West Bengal (Kolkata-based publications), so that the corpus is not biased toward the dialect of either region.

3.2 OCR Pipeline for Historical Documents

Bangla OCR is substantially harder than OCR for Latin-script languages because the script uses complex ligatures (*juktakkhar*), multiple diacritical marks that modify consonant glyphs, and variable ink density in historical print. Standard off-the-shelf OCR tools produce unacceptably high error rates on such material without preprocessing.

To mitigate this, we implemented an enhanced OCR workflow with the following stages:

- (1) **Image Preprocessing:** Each scanned page is converted to greyscale, sharpened with an unsharp mask filter, and binarised using Otsu’s method to improve contrast between ink and paper background.
- (2) **Resolution Tuning:** We iteratively increased scan resolution from 400 to 500 to 600 dpi, retaining the setting (600 dpi) that produced the highest character-level accuracy on a held-out validation set.
- (3) **OCR Engine:** We used Tesseract with the Bangla language model (`ben.traineddata`).
- (4) **Post-Correction:** A rule-based correction layer addressed the most common systematic errors: character confusion pairs identified empirically (e.g., / confusion), random Latin-script intrusions, and tokenisation noise from split/merged words.

To estimate the residual error rate, we manually transcribed a stratified random sample of 50 pages across the three historical eras and compared them character-by-character with the OCR output. We observed a mean Character Error Rate (CER) of approximately 20% across all historical eras, with Era 1 showing the highest rate (~25%) owing to poorer ink quality in early prints. Although this is a significant error rate, it is consistent with reported CERs for other historical non-Latin scripts [21], and our downstream preprocessing pipeline is designed to be robust to such noise through frequency thresholding (described in Section 4.2).

3.3 Preprocessing Pipeline

After collection and OCR, all raw files were processed through a custom cleaning pipeline implemented in Python:

- (1) **Unicode Normalisation:** We applied NFC normalisation to standardise the encoding of Bangla conjunct consonants, which can be represented in multiple Unicode-equivalent ways depending on the input source.
- (2) **Noise Removal:** We removed digits, Latin-script text, non-Bangla punctuation, and repeated-character noise sequences introduced by OCR.
- (3) **Token Filtering:** We retained only tokens that pass a Bangla phonotactic heuristic—specifically, a minimum token length of 2 characters and the presence of at least one Bangla vowel or vowel diacritic—to prune garbled OCR tokens that do not correspond to valid words.
- (4) **Sentence Segmentation:** Sentence boundaries were marked after each *dari* (।) character, the standard sentence-final punctuation mark in Bangla, to define clear context windows for embedding training.

Figure 1 illustrates the full preprocessing pipeline.

Fig. 1. The end-to-end preprocessing pipeline, from raw scanned text to clean tokenised output suitable for both n-gram analysis and embedding training.

Table 1. Corpus statistics across the four eras after preprocessing. “Cleaned Tokens” are the tokens surviving the Bangla phonotactic filter described in Section 3.

Era	Raw Tokens	Cleaned Tokens	Lines	CER (approx.)
1950–1970	940,652	394,833	78,495	~25%
1970–1990	916,711	398,296	71,510	~22%
1990–2010	1,109,531	407,820	53,852	~15%
2010–2025	1,296,724	739,623	93,971	0%

Fig. 2. Method overview. Raw diachronic text feeds two parallel modules: (A) the N-gram Module, which computes frequency trajectories and PPMI collocate sets per era; and (B) the Embedding Module, which trains per-era Word2Vec models and aligns them via Orthogonal Procrustes. Both modules feed a Fusion and Ranking stage that produces the final ranked list of semantically dynamic words.

3.4 Corpus Statistics

Table 1 summarises the size of each era after preprocessing. The cleaned token counts reflect the impact of the noise-removal pipeline: OCR-noisy historical eras lose a higher fraction of raw tokens (approximately 58% in Era 1) compared to the clean modern era (approximately 43% in Era 4). The modern era is also the largest by a substantial margin, which is typical of diachronic corpora assembled from heterogeneous sources and is a known confound that we address through per-million normalisation in all frequency analyses.

We acknowledge that the cleaned per-era token counts in the range of 400K–740K tokens are substantially smaller than the corpora used in high-resource LSC studies (Hamilton et al. used corpora of up to 8.5×10^{11} tokens). This limitation is inherent to the historical Bangla data situation and motivates our explicit robustness measures: longer training epochs for smaller eras, frequency thresholding on embeddings, and caution in interpreting results for rare words.

4 Methodology

Our analysis pipeline is designed to answer two distinct but complementary questions: (i) *which words are entering the language for the first time or surging in usage* (lexical emergence), and (ii) *which existing words are using their contexts in a fundamentally different way* (meaning shift). The first question is best addressed by transparent frequency-based signals; the second by distributional vector algebra. Figure 2 illustrates the two-module architecture.

4.1 Problem Formalisation

Let $\mathcal{C}$ be the diachronic corpus partitioned into four sequential eras $E = \{e_1, e_2, e_3, e_4\}$, where e_1 represents 1950–1970 and e_4 represents 2010–2025. For any target word $w \in \mathcal{V}$ and era $e \in E$, let C_e denote the tokenised corpus of era e , $|C_e|$ its token count, and $\mathcal{V}_e$ its vocabulary. Our objective is to compute metrics that capture two distinct phenomena for each word:

- (1) **Meaning Shift (“Chameleons”)**: The gradual drift in the semantic neighbourhood of an existing word, measured by comparing its distributional context between the earliest and latest eras.
- (2) **Lexical Emergence (“Spikers”)**: The rise of neologisms and loanwords that are absent from or negligible in the early era but prominent in the modern era, measured by frequency growth.

4.2 N-gram Module: Capturing Frequency and Collocate Signals

Frequency Trajectories. To track how the usage rate of a word evolves across eras of varying size, we compute its smoothed *frequency per million tokens* (freqPM) using additive (Laplace) smoothing:

$$\text{freqPM}(w, e) = 10^6 \cdot \frac{\text{count}(w \in C_e) + \alpha}{|C_e| + \alpha|\mathcal{V}_e|}, \quad \alpha = 1. \quad (1)$$

Additive smoothing with $\alpha = 1$ is necessary to avoid zero frequencies for words absent in small historical eras, and normalisation by per-million tokens makes frequency values comparable across eras of different sizes. A word whose freqPM increases dramatically from e_1 to e_4 is a candidate “Spiker”—a word entering or surging in the language, typically associated with a new concept or loanword.

PPMI Collocate Analysis. Frequency trajectories alone cannot reveal *why* a word’s usage changed: a word may surge in frequency while retaining exactly the same meaning (e.g., because a new event makes its referent more salient), or it may change meaning without any frequency change. To detect contextual change, we extract the collocate set of each target word from a symmetric window of $W = 5$ tokens on each side and measure the association strength between a word w and each collocate c using *Positive Pointwise Mutual Information* (PPMI) [5]:

$$\text{PPMI}(w, c) = \max\left(0, \log \frac{P_e(w, c)}{P_e(w) P_e(c)}\right). \quad (2)$$

PPMI measures how much more often w and c co-occur than expected under statistical independence. Values are clipped at zero to discard negative associations, which are statistically unreliable [10]. We retain only collocates that co-occur at least $k = 3$ times with w in the modern era to reduce noise from OCR-induced spurious tokens.

We represent the collocate structure of each word per era as its top-20 PPMI collocate set. The *Collocate Turnover* score for word w between eras e_1 and e_4 is then the Jaccard distance between these sets:

$$\text{ColTurnover}(w) = 1 - \frac{|\text{Top}_{20}(w, e_1) \cap \text{Top}_{20}(w, e_4)|}{|\text{Top}_{20}(w, e_1) \cup \text{Top}_{20}(w, e_4)|}. \quad (3)$$

A ColTurnover of 1.0 indicates complete replacement of the word’s contextual neighbourhood across 75 years, which is strong *prima facie* evidence of semantic change. We visualise these dynamics as a *collocate heatmap*: a matrix that plots the top collocates from any era against all four time periods, using a colour scale (PPMI score) to show which associations appeared, strengthened, weakened, or vanished over time.

4.3 Embedding Module: Quantifying Latent Semantic Drift

Per-Era Word2Vec Training. To capture latent semantic relationships beyond simple co-occurrence counts, we train a separate Word2Vec Skip-gram model [17] on each era’s cleaned corpus. Skip-gram is trained to predict the context words surrounding a target word within a fixed window, learning dense, low-dimensional vector representations that encode semantic proximity as geometric closeness. We use the following hyperparameters, which follow established recommendations for diachronic studies [10, 23]: embedding dimension $d = 300$; context window size = 5; number

of negative samples = 10; minimum token frequency min_count = 2. To compensate for the smaller size of historical eras, we increase the number of training epochs for eras e_1 and e_2 , which stabilises the embeddings by providing more gradient updates per token.

Orthogonal Procrustes Alignment. Independently trained Skip-gram models inhabit arbitrarily oriented coordinate systems: the same word in two different models will generally have different vector coordinates even if its meaning has not changed, simply because stochastic gradient descent converges to a different (but equally valid) rotational arrangement. Direct cosine-distance comparison across models is therefore meaningless without first aligning them.

We resolve this by solving the *Orthogonal Procrustes* problem [26]: given embedding matrices $\mathbf{A}$ (era e_1 reference) and $\mathbf{B}$ (era e , to be aligned), find the orthogonal rotation matrix $\mathbf{R}^*$ that best maps $\mathbf{B}$ onto $\mathbf{A}$ in the least-squares sense:

$$\mathbf{R}^* = \arg \min_{\mathbf{R}: \mathbf{R}^\top \mathbf{R} = \mathbf{I}} \|\mathbf{B}\mathbf{R} - \mathbf{A}\|_F, \quad (4)$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ enforces orthogonality, preserving all pairwise cosine similarities within each era. The matrices $\mathbf{A}$ and $\mathbf{B}$ are formed by stacking the vectors of a shared set of *high-frequency anchor words* that are present in all eras and expected to be semantically stable (e.g., common function words and body-part nouns). The solution is obtained efficiently via SVD and all later-era vectors are subsequently transformed as $\mathbf{v} \mapsto \mathbf{v}\mathbf{R}^*$, making them directly comparable to era- e_1 vectors.

Drift Metrics. After alignment, we compute two complementary drift scores for each word w present in both the earliest (e_1) and latest (e_4) eras:

- **Self-similarity drop** $\Delta_{\cos}(w) = 1 - \cos(\hat{\mathbf{v}}_w^{(e_1)}, \hat{\mathbf{v}}_w^{(e_4)})$, where $\hat{\mathbf{v}}$ denotes the ℓ_2 -normalised aligned vector. A high self-similarity drop means the word has moved far in the embedding space, indicating that its distributional context has changed substantially.
- **Neighbour overlap** $\text{Jaccard}@K(w) = \frac{|N_K^{e_1}(w) \cap N_K^{e_4}(w)|}{|N_K^{e_1}(w) \cup N_K^{e_4}(w)|}$, where $N_K^e(w)$ denotes the set of the $K = 50$ nearest neighbours of w in era e 's aligned embedding space. This captures whether the words that are most semantically associated with w have changed, even when the word itself moves only slightly—an important signal for words that add a new sense without fully losing their old one.

4.4 Unified ChangeScore and Ranking

To synthesise both drift metrics into a single score on a common scale, we compute the *ChangeScore* as the normalised sum of the self-similarity drop and the neighbour distance:

$$\text{CS}(w) = \frac{\Delta_{\cos}(w)}{2} + (1 - \text{Jaccard}@50(w)). \quad (5)$$

Dividing $\Delta_{\cos}$ by 2 scales it to the $[0, 1]$ range (its theoretical maximum is 2, since cosine distance $\in [0, 2]$ for ℓ_2 -normalised vectors), matching the $[0, 1]$ range of the Jaccard distance component. The ChangeScore thus ranges from 0 (no change) to 2 (complete semantic replacement).

For “Spiker” words that are absent from e_1 but prominent in e_4 , no self-similarity drop can be computed; they are ranked separately by their $\text{freqPM}(w, e_4)$ score. The final reported ranking interleaves the top-ranked Chameleons (by ChangeScore) and the top-ranked Spikers (by freqPM), creating a comprehensive map of lexical dynamics.

Fig. 3. Distribution of ChangeScores across the shared vocabulary (words present in both 1950–1970 and 2010–2025 eras). The long right tail identifies words with high semantic drift; the majority of words cluster near the mean, indicating broad lexical stability.

Table 2. Top 5 most dynamically changing and most stable words in the shared vocabulary, ranked by ChangeScore (1950–1970 vs. 2010–2025).

Most Changed		Most Stable	
Word	Score	Word	Score
	1.912		1.249
	1.911		1.265
	1.910		1.319
	1.904		1.322
	1.901		1.324

4.5 Testing the Laws of Semantic Change

To situate our Bangla findings within the broader theoretical framework of cross-linguistic semantic change, we test two statistical laws proposed by Hamilton et al. [10]:

- (1) **Law of Frequency:** The rate of semantic change scales as a negative power law of word frequency—frequent words change more slowly. We test this by computing the Spearman rank correlation ρ between $\log_{10}(\text{freqPM}(w, e_1))$ and $\text{CS}(w)$ across all words in the shared vocabulary.
- (2) **Law of Polysemy:** After controlling for frequency, polysemous words (those with multiple senses) change faster. Because a hand-curated sense inventory does not exist for Bangla, we use *neighbour dispersion*—the average pairwise cosine distance among a word’s $K = 50$ nearest neighbours in e_1 —as a computational proxy for polysemy. A word whose nearest neighbours are spread far apart in the vector space likely appears in diverse, unrelated contexts, suggesting multiple senses [10].

5 Results

5.1 Overall Distribution of Semantic Change

Figure 3 presents the distribution of ChangeScores across all words present in both e_1 and e_4 . The distribution is strongly right-skewed: the bulk of shared vocabulary clusters around a mean score of approximately 1.62, indicating that the majority of words in the Bangla lexicon are semantically stable over 75 years. However, the long right tail—words with ChangeScores above ~ 1.85 —identifies a key set of existing words that have undergone substantial semantic drift. This distribution shape is consistent with LSC findings in other languages [24], providing a first sign that Bangla follows cross-linguistic patterns of lexical stability.

Table 2 presents the five most semantically dynamic words (highest ChangeScore) and the five most stable words (lowest ChangeScore).

5.2 Frequency Trajectories of Showcase Words

Figure 4 presents the smoothed freqPM trajectories of three showcase words selected to exemplify the three dominant patterns of change identified in our data. The trajectories visually confirm the expected signatures: a flat then explosive rise for the lexical-emergence word (), a distinct spike in the 1970–1990 period followed by renewed growth for the

Fig. 4. Smoothed frequency per million tokens (freqPM) for three showcase words across the four eras, illustrating three qualitatively distinct patterns of change: lexical emergence (XXXXXX), historical anchoring (XXXXXXXXXX), and meaning shift (XX).

(a) Collocate heatmap for XXXXXX. Green cells indicate high PPMI association; the heatmap shows the sudden appearance of technology-related collocates in e_3 and their replacement by social/commercial terms in e_4 . (b) 2D UMAP projection of XXXXXX's neighbourhood in e_3 (blue) vs. e_4 (red). The spatial separation of the two era clusters visualises semantic broadening from a technical to a societal concept.

Fig. 5. Collocate and embedding evidence for the lexical emergence of XXXXXX (*digital*).

historically anchored word (), and a steady decline for the meaning-shift word (), whose frequency drop is misleading because the dominant *sense* has changed.

5.3 Qualitative Case Studies

We now present detailed analyses of three words that exemplify different mechanisms of semantic change. For each, we synthesise evidence from frequency trajectories, collocate heatmaps, and embedding drift plots to build a coherent narrative of the word's semantic trajectory.

5.3.1 Case Study 1: XXXXXX (*digital*) — Lexical Emergence. The Bangla word (*dijital*) is a phonological borrowing of the English adjective *digital*. It provides the clearest possible illustration of *lexical emergence*: a word that does not exist in the early language but enters rapidly and subsequently undergoes semantic broadening as the concept it names becomes culturally central.

is absent from our corpus in eras e_1 and e_2 (1950–1990), consistent with the fact that the concept of digital computing had no presence in the everyday discourse of Bangla speakers of that period. Its freqPM rises steeply in era e_3 (1990–2010) as personal computers and the internet arrive, and then spikes dramatically in e_4 (2010–2025) as the word moves beyond technical discourse into political, commercial, and everyday registers.

Why this matters. Frequency alone does not tell us whether the word's *meaning* changed or merely its frequency of use. The collocate analysis is essential here: it shows that in e_3 the word's top PPMI collocates include (*oscillation*), (*electronic*), and (*computer*), defining a narrow, specialist technical meaning. By e_4 , these technical neighbours are replaced by (*platform*), (*marketing*), and (*commerce*), reflecting the word's broadening from a technical term to a general societal concept encompassing anything modern and technology-mediated. The collocate heatmap (Figure 5a) makes this transition visually unambiguous.

How the embedding confirms it. Figure 5b presents a 2D UMAP projection of the word's embedding neighbourhood in e_3 (blue) versus e_4 (red) after Procrustes alignment. The significant spatial displacement between the two era vectors, and the complete migration of the word's nearest-neighbour cluster from technical to societal terms, corroborates the ChangeScore and provides interpretable geometric evidence of semantic broadening.

5.3.2 Case Study 2: XX (*kul/cool*) — Meaning Shift. The word (*kul*) is a “chameleon” word: its frequency has declined over 75 years, which might suggest obsolescence, but its *meaning* has been entirely replaced rather than lost. This case demonstrates why frequency analysis alone is insufficient to characterise semantic change, and why embedding drift is essential.

(a) Collocate heatmap for বাংলা , showing the complete replacement of early collocates related to ‘lineage’ with modern collocates related to ‘style’ and ‘fashion’.

(b) 2D UMAP projection of বাংলা ’s neighbourhood in e_3 (blue) vs. e_4 (red). The near-complete disjointness of the two clusters confirms semantic replacement.

Fig. 6. Collocate and embedding evidence for the meaning shift of বাংলা .

What happened. The word has two distinct histories. In classical and formal literary Bangla (dominant in e_1 and e_2), it means *lineage*, *ancestry*, or socio-religious group membership—an abstract concept of social hierarchy. In contemporary informal urban Bangla (dominant in e_4), it is used as a transliteration of the English *cool*, meaning fashionable, stylish, or socially admired. The word’s declining freqPM reflects the decay of the formal *lineage* meaning faster than the informal *cool* meaning rises in the written corpus.

Why this matters. This case illustrates the phenomenon of *semantic replacement* driven by cultural globalisation: an existing phonological form is repurposed to express a concept imported from a dominant global language. Similar replacements have been observed for English-origin words entering French and German [29], but to our knowledge this specific mechanism has not been documented for Bangla. The collocate heatmap (Figure 6a) makes the replacement visible: collocates related to (*lineage/family*) dominate e_1 – e_2 and vanish completely by e_4 , replaced by (*fashionable*) and (*stylish*).

How the embedding confirms it. Figure 6b shows the 2D UMAP projection of বাংলা ’s neighbourhood between e_3 and e_4 . The two neighbourhood clusters are almost entirely disjoint, confirming that the word’s modern semantic neighbourhood has no overlap with its transitional-era neighbourhood—a geometric signature of semantic replacement rather than semantic drift.

5.3.3 Case Study 3: স্বাধীনতা (swadhinata/independence) — Historical Anchoring. The word (*swadhinata*) demonstrates a third mechanism of semantic change that is qualitatively distinct from the two above: *historical anchoring*, in which a word’s core denotation remains unchanged but its connotational and collocational field is profoundly reshaped by a single historical event.

What happened. Before 1971, স্বাধীনতা was a general abstract noun meaning *freedom* or *independence*, used in contexts ranging from personal autonomy to colonial resistance. After the Bangladesh Liberation War of 1971, the word acquired a specific, institutionalised meaning: the independence of *Bangladesh*, which became its dominant, culturally foregrounded sense. Our frequency data (Figure 4) shows a distinct surge in the 1970–1990 era coinciding with post-independence nation-building discourse, followed by a larger spike in e_4 as the word becomes central to digital-age national commemoration.

Why this matters. Historical anchoring is theoretically important because it violates the assumption of smooth continuous semantic drift that underpins most LSC models [14]: the change is abrupt, externally caused, and partially intentional (state institutions actively promote specific commemorative uses of the word). Detecting this type of change requires the collocate heatmap (Figure 7a), which shows the shift from (*struggle*) in e_1 to (*war*) in e_2 to (*day*, as in Independence Day) in e_4 .

(a) Collocate heatmap for স্বাধীনতা (independence), tracking the word’s context from general ‘struggle’ in e_1 , to specific ‘war’ in e_2 , to modern ‘commemoration’ in e_4 . (b) 2D UMAP neighbourhood drift for স্বাধীনতা between e_3 (blue) and e_4 (red), showing continued evolution toward a commemorative semantic field.

Fig. 7. Collocate and embedding evidence for the historical anchoring of স্বাধীনতা (independence).

Fig. 8. ChangeScore versus historical (1950–1970) log frequency for the shared vocabulary. The downward-sloping red trend line confirms the Law of Frequency holds for Bangla: more frequent words are more semantically stable. Spearman $\rho = -0.083$, $p < 0.001$.

Fig. 9. ChangeScore versus polysemy proxy (neighbour dispersion in 1950–1970). The flat trend line indicates no significant correlation ($\rho = -0.020$, $p = 0.087$), suggesting the Law of Polysemy is not supported under the present corpus conditions.

How the embedding confirms it. Figure 7b shows that the word continues to drift semantically even between e_3 and e_4 : the modern neighbourhood includes (great) and (victory), confirming that national memory is still actively reshaping this word’s semantic space in the digital era.

5.4 Testing the Statistical Laws of Semantic Change

5.4.1 Law of Frequency: Confirmed. Figure 8 presents the scatter plot of ChangeScore against historical log-frequency ($\log_{10}$ freqPM in e_1) for the entire shared vocabulary. A Spearman rank correlation analysis yields $\rho = -0.083$ with $p < 0.001$, confirming a statistically significant negative relationship: words that were more frequent in the early era tend to have lower ChangeScores, indicating greater semantic stability.

The effect size is weak ($|\rho| = 0.083$), which is consistent with findings in other languages where corpus size limits statistical power [10]. Nonetheless, the direction and significance of the correlation confirm that Bangla is not an exception to this cross-linguistic regularity. This result has a practical implication for future Bangla NLP: common function words and high-frequency nouns form a stable semantic core that can be reliably modelled using contemporary corpora even when processing historical documents.

5.4.2 Law of Polysemy: Not Confirmed under Current Corpus Conditions. Figure 9 presents the scatter plot of ChangeScore against our polysemy proxy (neighbour dispersion in e_1). A Spearman rank correlation yields $\rho = -0.020$ with $p = 0.087$ —a non-significant result. The flat trend line confirms that our data provide no evidence for the Law of Polysemy under current corpus conditions.

We argue that this null result should not be interpreted as evidence that Bangla violates the Law of Polysemy. The neighbour dispersion proxy is highly sensitive to word vector quality, which degrades with corpus size. Our historical eras contain approximately 400K cleaned tokens each—roughly two orders of magnitude smaller than the corpora used by Hamilton et al. to confirm the law. In sparse corpora, rare words are represented by noisy vectors whose nearest-neighbour sets are unstable, which would degrade the polysemy proxy and obscure any underlying relationship. This interpretation is consistent with the findings of Dubossarsky et al. [7], who showed that polysemy-change correlations are model-sensitive, and with similar null results reported for Romanian [31].

6 Discussion

6.1 Interpreting the Semantic Changes in Bangla

Our results map three qualitatively distinct mechanisms through which the Bangla lexicon has evolved between 1950 and 2025, and these mechanisms correspond to identifiable extra-linguistic forces.

Technological adoption drives lexical emergence. The “Spiker” pattern observed for is a canonical example of how technological innovation generates new vocabulary: a foreign term enters the language as a borrowing, initially restricted to specialist registers, and progressively expands into general discourse as the underlying technology permeates daily life. The semantic broadening we observe—from “related to digital circuits” to “modern, tech-savvy, online”—mirrors similar trajectories documented for *broadcast* in English (from “scattering seeds” to “transmitting radio signals”, Hamilton et al. 10) and demonstrates that Bangla is not exceptional in this respect.

Cultural globalisation drives semantic replacement. The case is perhaps the most theoretically interesting because it involves the repurposing of an existing phonological form to carry an imported meaning. This process—sometimes called *semantic calquing* or *faux-ami borrowing*—is enabled by the phonological similarity between the Bangla word *kul* and the English *cool*, allowing the new sense to “ride” on the existing word’s phonology. The completeness of the replacement (near-zero collocate overlap between e_1 and e_4) suggests that the older *lineage* sense has been substantially marginalised in everyday written Bangla, at least in the journalistic and online registers that dominate our modern corpus.

National history anchors meaning. The case illustrates that words can be deliberately *institutionalised* around specific historical events, creating what might be called a *commemorative semantic field* that resists subsequent drift. The continued evolution we observe between e_3 and e_4 —toward (*great*) and (*victory*)—suggests that this anchoring is not static but is continuously reinforced by political discourse, annual commemorations, and digital media, which amplify nationalist vocabulary in contemporary Bangla.

6.2 Methodological Contributions and Lessons for Low-Resource Diachronic NLP

Our work demonstrates that it is possible to extract statistically reliable LSC signals from small, OCR-noisy corpora, provided that the pipeline is explicitly designed to mitigate noise at every stage. The key methodological decisions that enabled this are: (i) frequency thresholding to exclude OCR-corrupted hapax legomena from the embedding vocabulary; (ii) increased training epochs for small historical eras to compensate for reduced data volume; (iii) the use of a shared high-frequency anchor set for Procrustes alignment rather than the full vocabulary, which reduces the influence of rare, potentially misspelled tokens on the alignment; and (iv) the PPMI collocate heatmap as a qualitative sanity check that makes OCR-induced noise visible (it tends to produce collocates that are clearly not meaningful Bangla words, which can be identified and filtered).

These lessons are transferable to other South Asian languages with similar historical archive conditions, including Odia, Assamese, and Punjabi, for which the same combination of rich literary heritage and absent diachronic digital infrastructure applies.

6.3 Limitations

We acknowledge four limitations that constrain the scope of our current findings.

Corpus size. Our per-era cleaned token counts (~400K–740K) are small by LSC standards. This limits embedding quality, particularly for rare words, and is the most likely explanation for the null polysemy law result. Future

work should prioritise corpus expansion through additional newspaper archives, library digitisation projects, and audio/video transcript collections from Bangla broadcast media.

Genre imbalance. Our corpus blends formal journalistic text with literary prose, and the modern era additionally includes web text. Genre shifts can mimic or mask semantic shifts: a word whose usage expands from formal to informal registers will show collocate turnover that reflects register differences as much as meaning differences. Future work should construct genre-matched sub-corpora to disentangle these effects.

Sense conflation in static embeddings. Static Word2Vec models assign a single vector per word, averaging across all its senses. This is particularly problematic for polysemous words, where an old sense decays as a new sense rises. The PPMI collocate analysis partially mitigates this by operating at the level of individual collocate associations rather than aggregate vectors, but a fully satisfactory solution requires sense-level representations. BanglaBERT [3] offers a path forward once per-era fine-tuning becomes feasible with larger corpora.

Absence of a gold standard. Unlike the SemEval-2020 [24] and DWUG [25] resources for English, German, Latin, and Swedish, no human-annotated LSC benchmark exists for Bangla. Our evaluation therefore relies on qualitative case studies and the statistical law tests. Creating such a benchmark—even a small one covering 50–100 words with native-speaker annotations—would be a high-impact resource contribution for the community.

7 Conclusion

This paper presented the first large-scale computational study of lexical semantic change in Bangla across a pivotal 75-year window (1950–2025). We addressed two foundational challenges—the absence of a time-stratified corpus and the high OCR noise in historical archives—by constructing a four-era diachronic corpus and an error-aware preprocessing pipeline that retains sufficient signal quality for downstream analysis. Our hybrid analysis pipeline, combining PPMI collocate analysis with Procrustes-aligned Word2Vec embeddings and a unified ChangeScore, produced the first data-driven map of Bangla lexical evolution.

Our principal findings are threefold. First, we identified three qualitatively distinct mechanisms of semantic change in Bangla: lexical emergence driven by technological adoption, semantic replacement driven by cultural globalisation, and meaning anchoring shaped by national history—each evidenced by converging n-gram and embedding signals. Second, we confirmed that the Law of Frequency holds for Bangla ($\rho = -0.083$, $p < 0.001$): high-frequency words form a semantically stable core of the lexicon across our study period. Third, we found no evidence for the Law of Polysemy under our current corpus conditions, an outcome we attribute to data sparsity rather than to a genuine linguistic exception, and which motivates corpus expansion as the most critical next step.

Beyond the specific findings, this work establishes a reproducible baseline pipeline for diachronic analysis that is explicitly designed for low-resource, high-noise conditions. The diachronic corpus, the preprocessing scripts, and the analysis code are released publicly to support future work on Bangla and other under-resourced Indo-Aryan languages whose historical semantic evolution remains computationally unexplored.

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